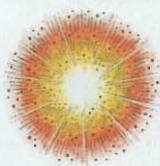


BIG BANG



AN INCREDIBLE EXPLOSION called the Big Bang is believed to have created the Universe. Observations of galaxies and heat radiation from space have helped confirm this theory. Astronomers are now working to explain exactly what happened from the point of the Big Bang explosion which created everything in today's Universe – matter, energy, space, and time – to the present Universe with its galaxies, stars, planets, and us.

Steady State theory

In the late 1940s and the 1950s, the Steady State theory was as popular as the Big Bang theory. It proposed that the Universe looked the same at any place and at any time. Although expanding, it would stay unchanged and in perfect balance. Material was being continuously created to keep the density of the Universe constant. As scientists found proof for the Big Bang, the Steady State theory was largely abandoned.



A Steady-State universe now (left) and later in time (right). The galaxies have moved apart, but new ones (coded orange) have been created to take their place. The density stays the same.

Georges Lemaître

In 1931, Belgian cosmologist Georges Lemaître (1894–1966) was the first to put forward the theory that the Universe started from a dense, single unit of material in a big explosion. The name Big Bang followed in 1950, introduced by Fred Hoyle, a British astronomer and supporter of the Steady State theory.



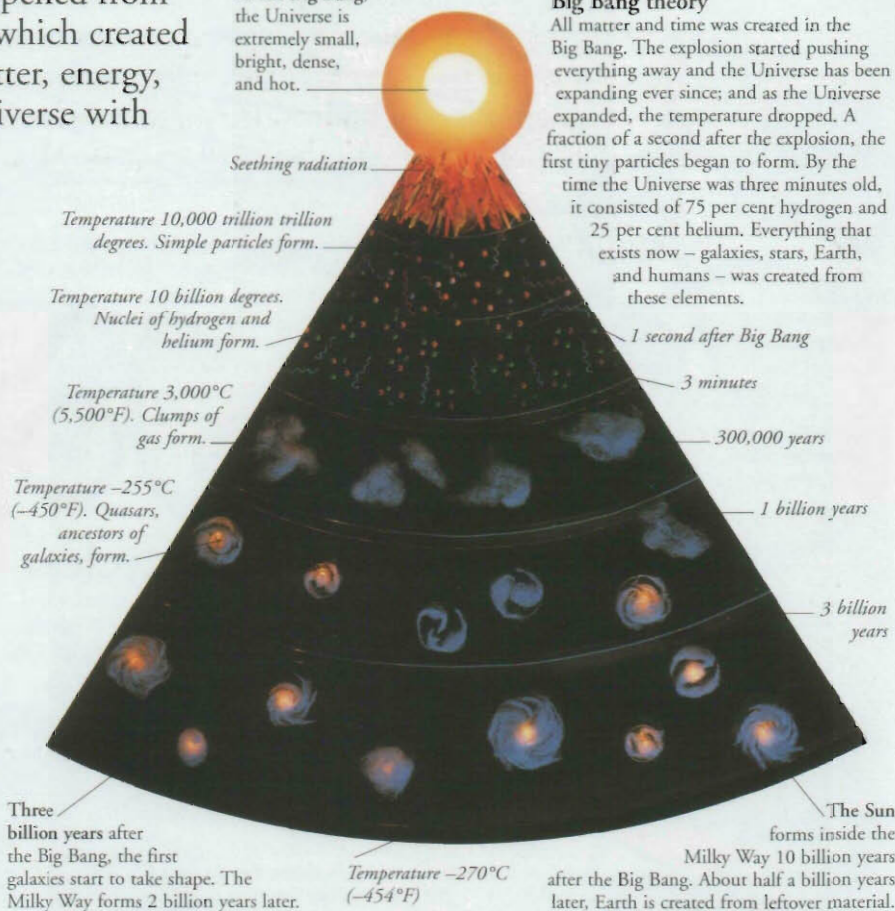
Origin of the Universe

One of the most difficult problems facing scientists in the 20th century was to explain how the Universe was created. The Universe is changing, but from what and to what? The Steady State theory suggested that the Universe had no beginning or end. The alternative, and now generally accepted, theory is the Big Bang. It proposes that the Universe was created in an explosion 15 billion years ago. From very small and simple beginnings it has grown vast and complex.

At the Big Bang, the Universe is extremely small, bright, dense, and hot.

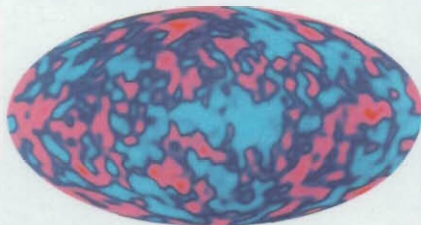
Big Bang theory

All matter and time was created in the Big Bang. The explosion started pushing everything away and the Universe has been expanding ever since; and as the Universe expanded, the temperature dropped. A fraction of a second after the explosion, the first tiny particles began to form. By the time the Universe was three minutes old, it consisted of 75 per cent hydrogen and 25 per cent helium. Everything that exists now – galaxies, stars, Earth, and humans – was created from these elements.



Expanding Universe

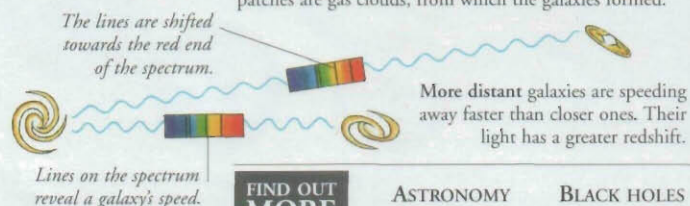
In the 1920s, analysing starlight from galaxies showed that the galaxies are moving away from Earth. This is true of galaxies in every direction from Earth. Over time, the Universe is becoming larger and less dense. The idea that the Universe started in an explosion from a single point grew out of observations that the Universe is expanding.



Background radiation

The heat produced by the Big Bang has been cooling ever since. It now has a temperature of -270°C (-454°F), detected as microwave radiation from all over the sky. The false-colour map shows variations in the temperature 300,000 years after the Big Bang. The blue (cooler) patches are gas clouds, from which the galaxies formed.

Redshift: The faster a galaxy is moving away, the more the wavelength of its starlight is stretched, or redshifted.

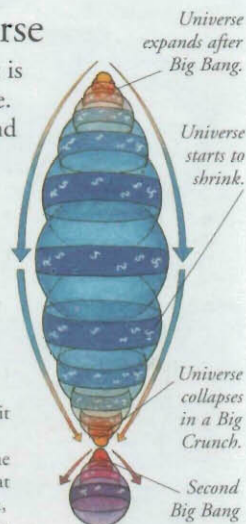


Future of the Universe

Nobody knows for certain what is going to happen to the Universe. At present, it is getting larger and less dense. Most astronomers believe there will be a time when it stops expanding. But there is disagreement about what happens then: will the Universe live on for ever, wither and die, or start to contract?

Big Crunch

The Universe may end in a Big Crunch if it starts to contract until it is hot and dense once more. But even this may not mean the end of the Universe. The Big Crunch might be followed by another Big Bang explosion, and the whole process could start over again.



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